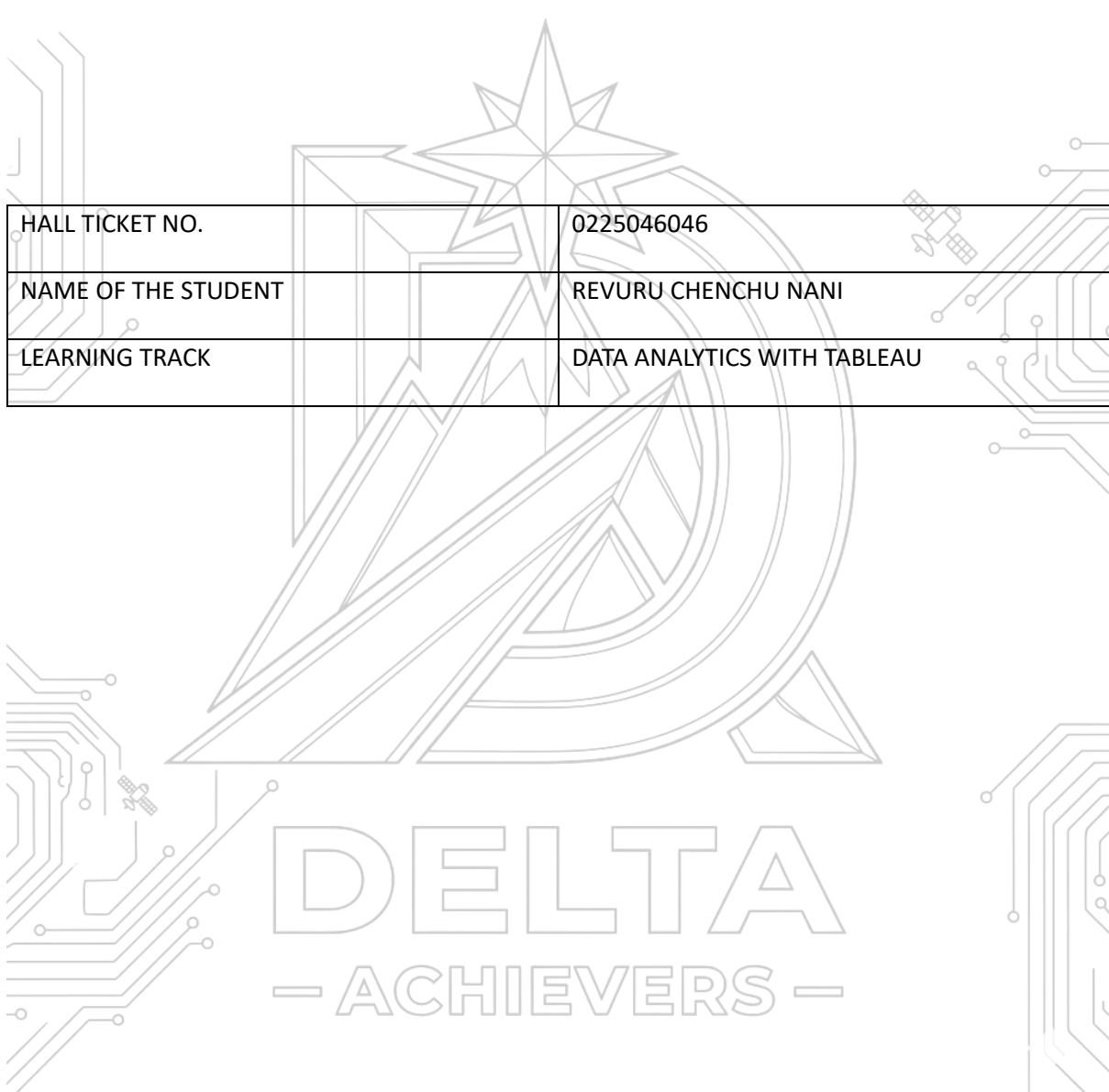


ASSIGNMENT

DATA ANALYTICS WITH TABLEAU

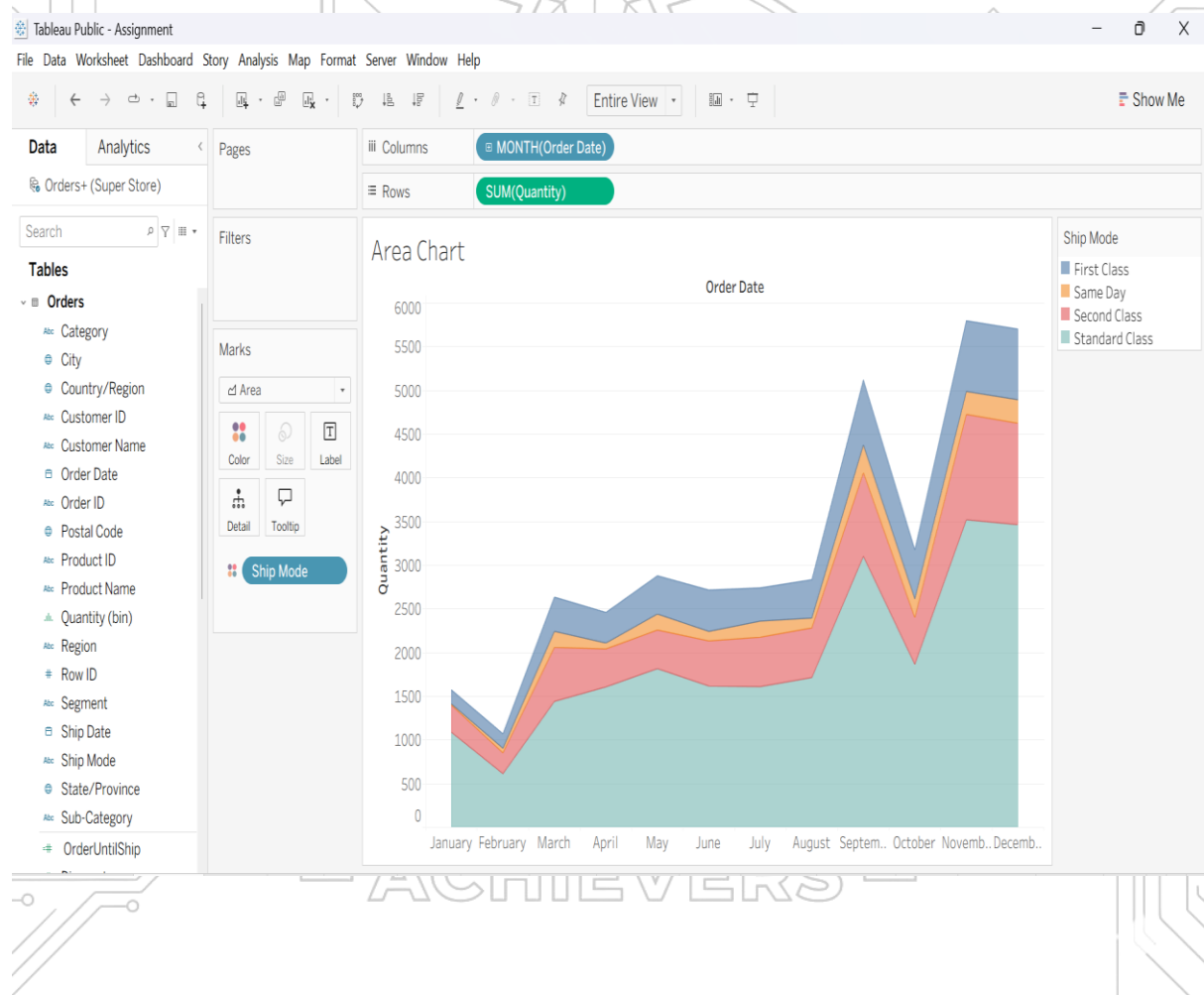
The background of the page features a large, faint watermark of the Delta Achievers logo, which includes a stylized star and the text 'DELTA - ACHIEVERS -'. The page is also decorated with circuit-like line art patterns on the left and right sides.

HALL TICKET NO.	0225046046
NAME OF THE STUDENT	REVURU CHENCHU NANI
LEARNING TRACK	DATA ANALYTICS WITH TABLEAU

Sheet – 01 – Area Chart Steps:

1. Drag Order Date to Columns Shelf – puts time on the x-axis.
2. Right-click YEAR(Order Date) → Select Month – changes the view from yearly to monthly breakdown. So you get Jan, Feb, Mar... on the x-axis.
3. Drag Quantity to Rows Shelf* – puts total quantity on the y-axis. Tableau auto-aggregates it to SUM(Quantity).
4. Drag Ship Mode to Color on Marks card – splits the area by ship mode and colors each segment: First Class, Same Day, Second Class, Standard Class.
5. Marks card → Drop-down → Select Area – changes the chart type from line/bar to area chart.

OUTPUT:



Sheet 02 – Bar Chart

1. Drag Sales to Rows Shelf

Tableau aggregates it as `SUM(Sales)` and defaults to a line chart since Order Date is usually on Columns.

2. Change to Bar Chart

On the Marks Card, open the drop-down and select -Bar.

3. Add Ship Mode Colors

Drag -Ship Mode to Color on the Marks card. This stacks the bar by First Class, Same Day, Second Class, Standard Class.

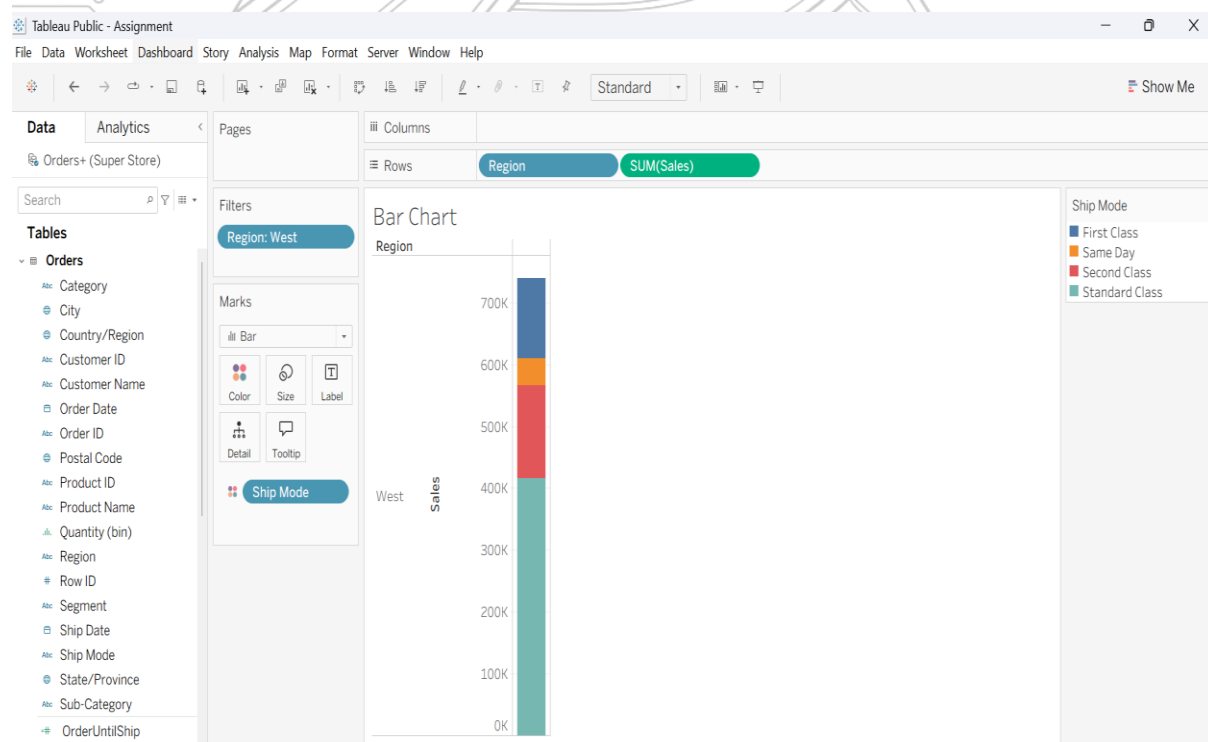
4. Add Region Comparison

Drag - Region to the Rows Shelf left of Sales. This creates a bar per region.

5. Filter to West Region Only

Drag Region to Filters Shelf → uncheck Central, East, South → keep only - West → click OK.

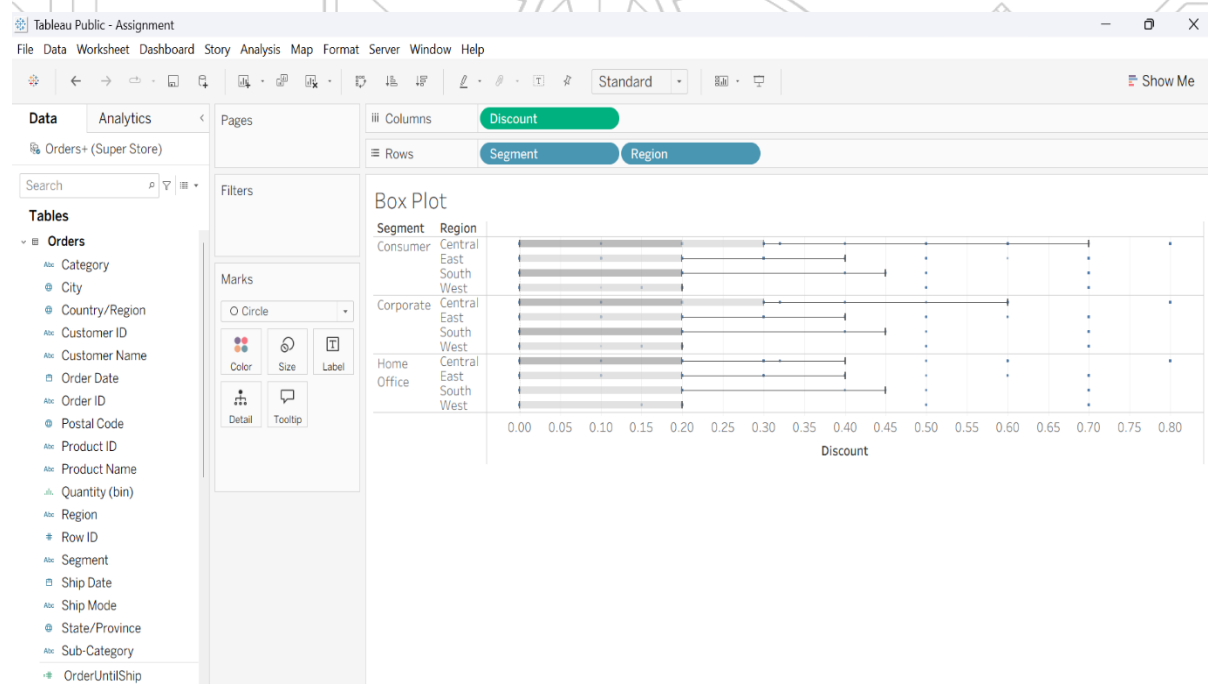
OUTPUT:



Sheet- 03 - Box Plot Steps:*

1. Drag - Segment to Columns and - Discount to Rows
2. Change Discount to -Average aggregation
3. Drag – Region to Columns, place it to the right of Segment
4. Click <Show Me > select -Box-and-Whisker Plot
5. Go to < Analysis menu > uncheck Aggregate Measures
6. Click Swap Rows and Columns for horizontal view

Output:

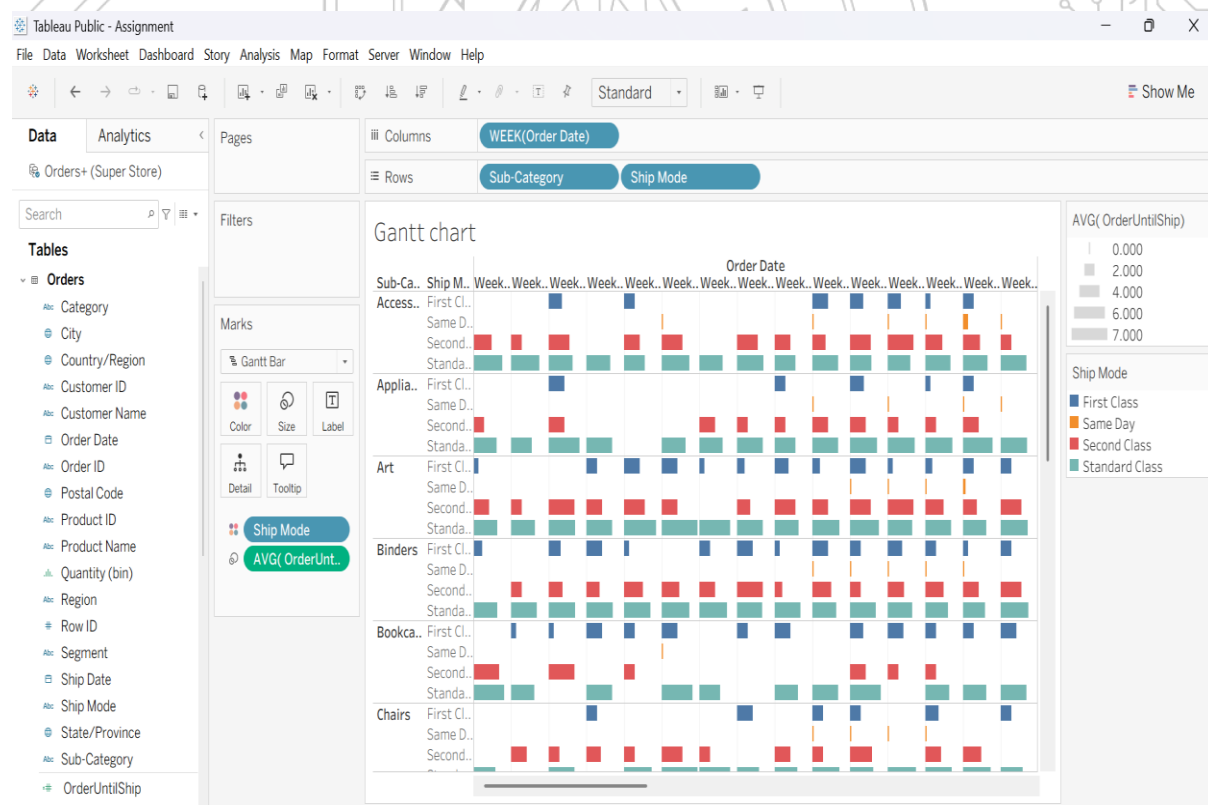


Sheet – 04 - Gantt Chart

Steps:

1. Drag ***Order Date*** to Columns > change to ***Week Number***
2. Drag ***Sub-Category*** and ***Ship Mode*** to Rows
3. On Marks card, change Mark Type to ***Gantt Bar***
4. Create calculated field: `OrderUntilShip = DATEDIFF('day', [Order Date], [Ship Date])`
5. Drag ***OrderUntilShip*** to ***Size*** on Marks card
6. Change it to ***Average*** by right-clicking > Measure > Average
7. Drag ***Ship Mode*** to ***Color*** on Marks card

Output:

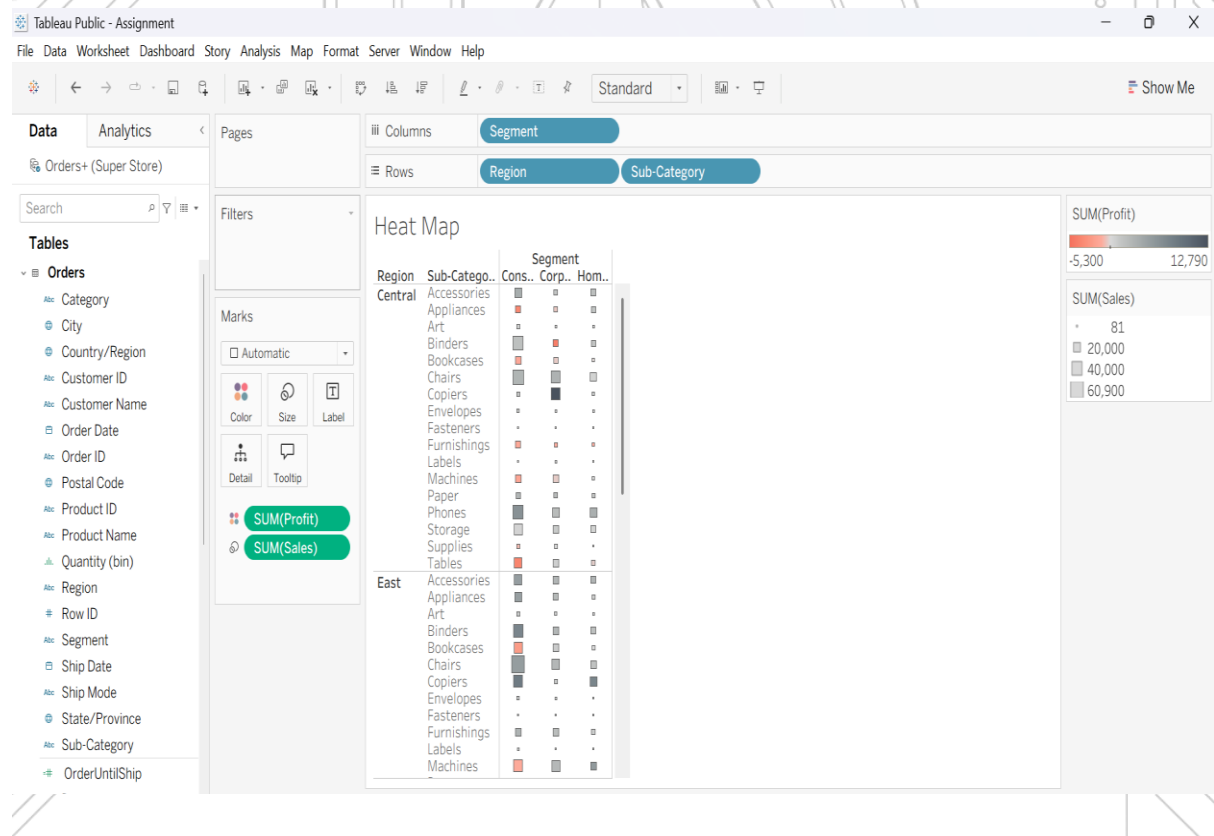


Sheet – 05 - Heat Map

Steps:

1. Drag – Segment to Columns
2. Drag - Region and - Sub-Category to Rows, with Sub-Category to the right of Region
3. Drag - Profit to - Color on the Marks card
4. Click <Color > <Border > choose - Medium Gray - to separate cells
5. Click the Profit color legend > Edit Colors select <Red-Black Diverging> palette > check Use Full Color Range > OK
6. Drag - Sales to - Size on the Marks card so bigger boxes = higher sales and smaller box = low sales
7. Increase box size click size on marks card slider to right boxes fit properly.

Output:

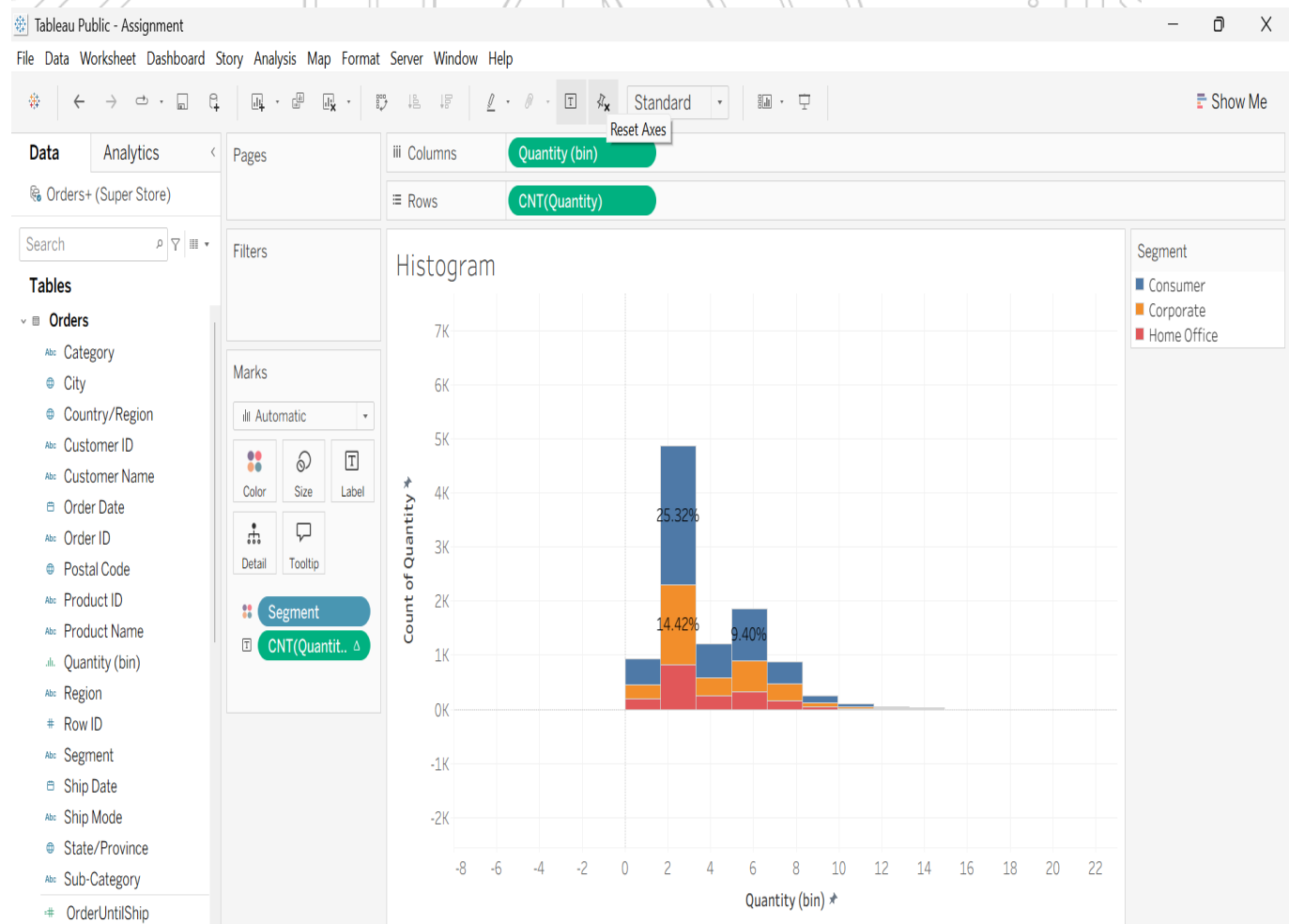


Sheet – 06 - Histogram

Steps:

1. Drag <Quantity> to Columns
2. Click <Show Me > select - Histogram
3. Drag – Segment- to Color to split bars by Consumer, Corporate, Home Office
4. Hold <Ctrl> and drag -CNT(Quantity)- from Rows to -Label- to show counts
5. Right-click -CNT(Quantity) on Marks card > Quick Table Calculation > Percent of Total to show percentages.

OUTPUT:

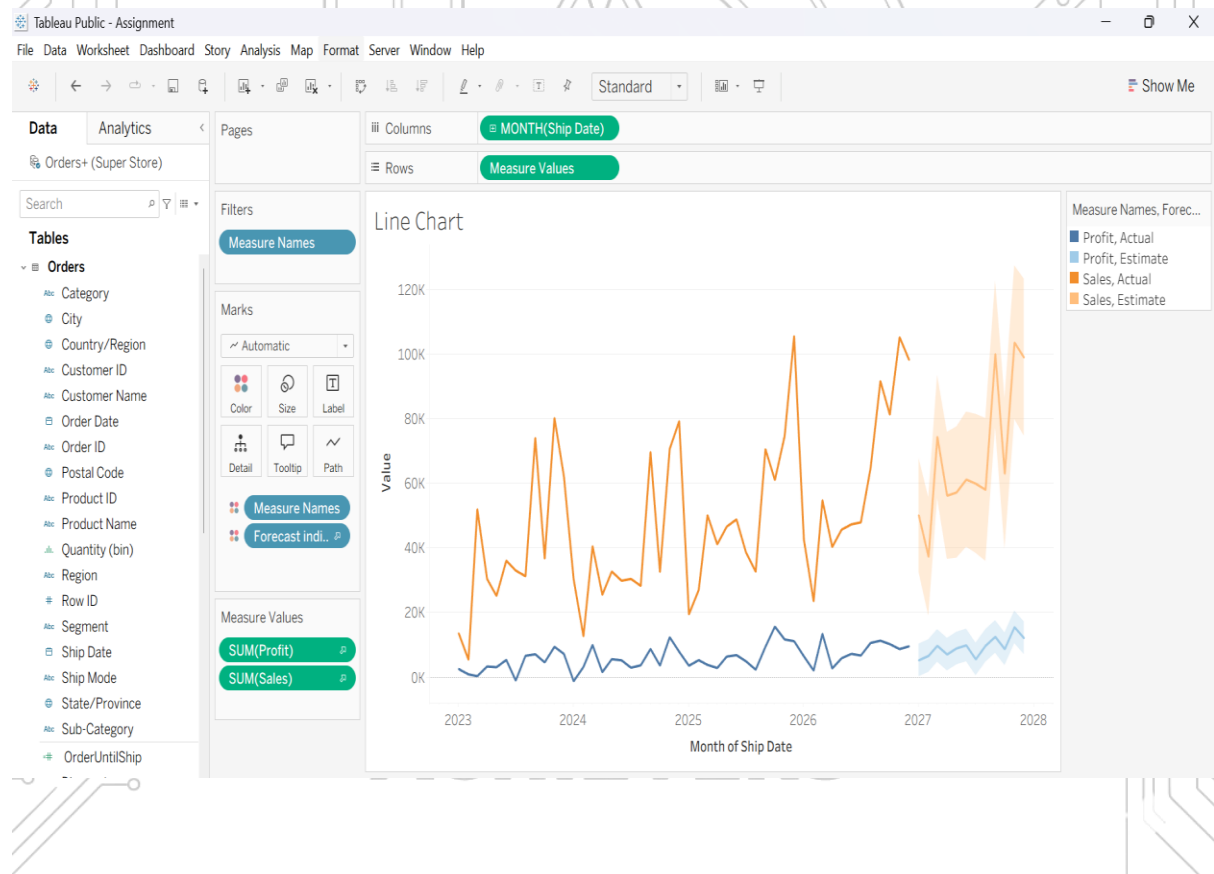


Sheet – 07 - Line Chart

Steps:

1. Drag -Ship Date to Columns → Tableau sets it to YEAR(Ship Date)
2. Drag – Sales- to Rows to create the line chart
3. Drag – Profit - to Rows, place it to the right of Sales → now you have 2 axes
4. Drag - SUM(Profit) onto the Sales axis to blend them onto one axis
5. Click the YEAR(Ship Date) dropdown > select Month (Continuous) for a monthly trend
6. Optional: Open the Analytics pane, drag Forecast into the chart to see predicted values.

OUTPUT:

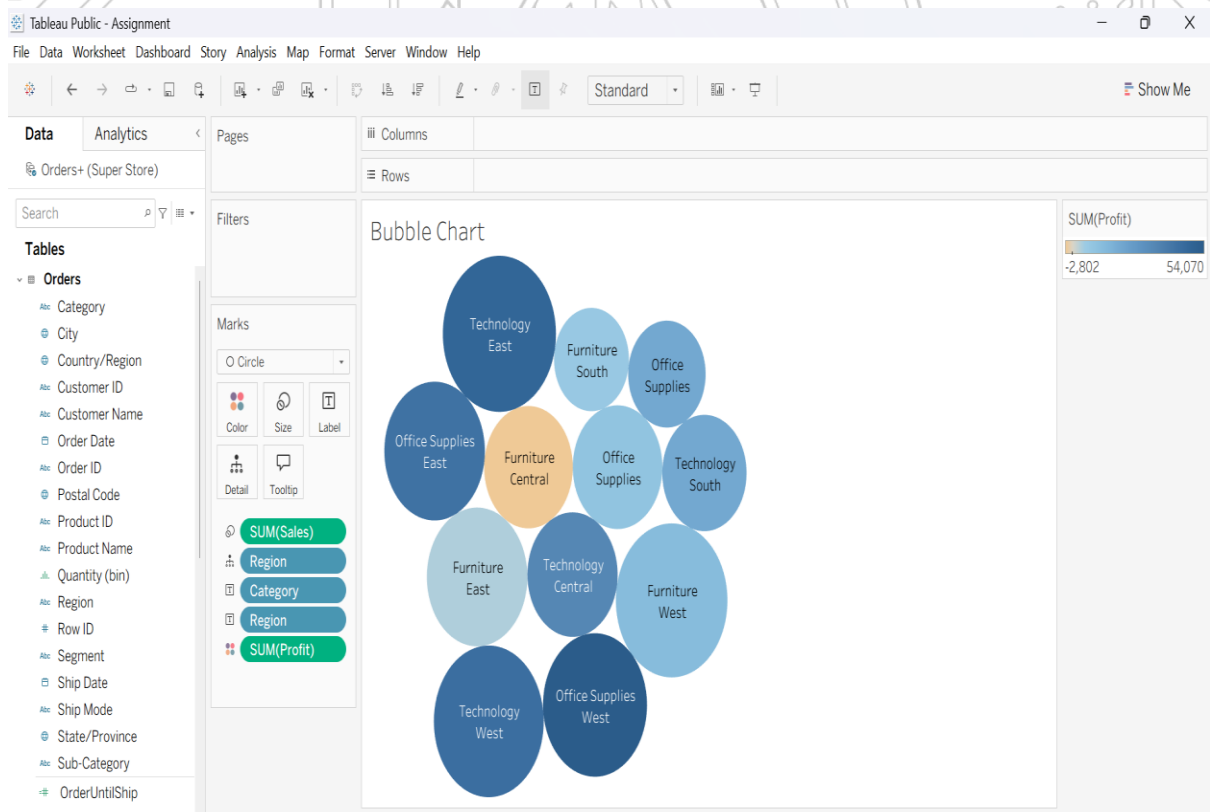


Sheet – 08 - Packed Bubble Chart:

Steps:

1. Drag Category to Columns
2. Drag - Sales- to Rows. Tableau will make a bar chart by default
3. Click <Show Me > select -Packed Bubbles. Bubble size = Sales
4. Drag <Region> to -Detail- on the Marks card to split each bubble by region
5. Drag <Profit> to - Color - on the Marks card. Color shows profit vs loss
6. Drag <Region> to – Label - on the Marks card to display region names on each bubble.

OUTPUT:

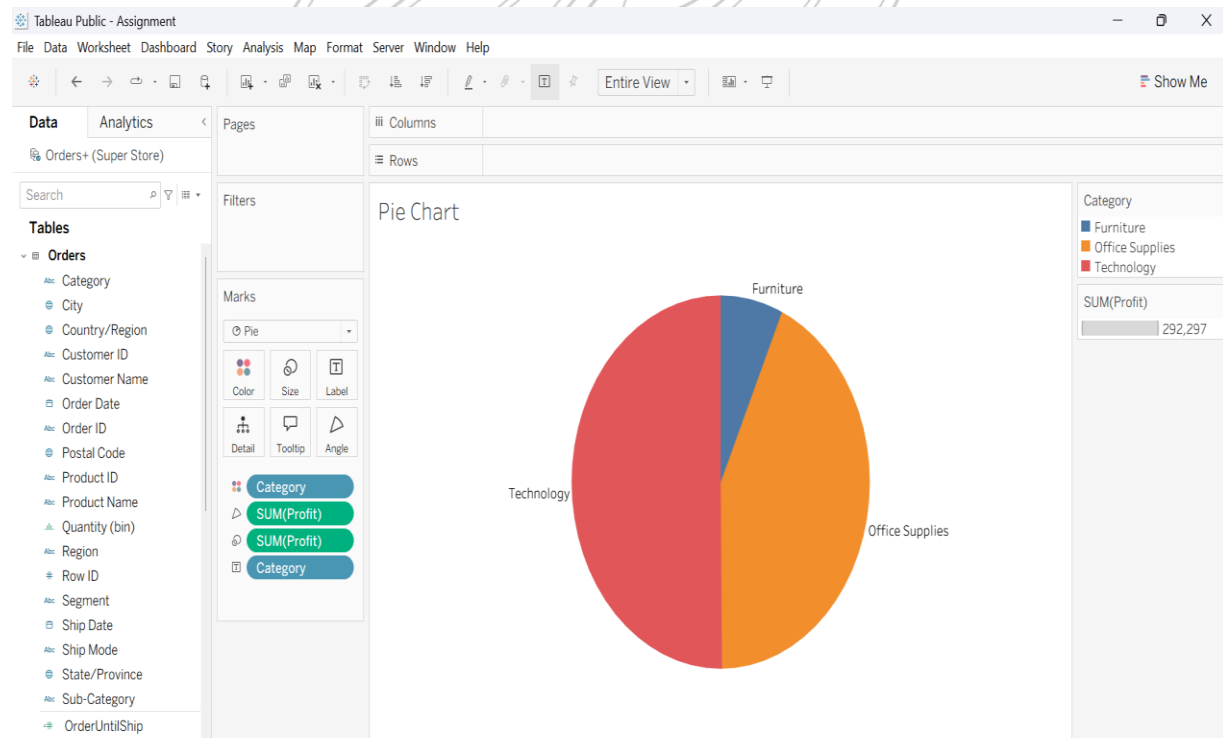


Sheet – 09 - Pie Chart:

Steps:

1. Drag <Profit> to Columns
2. Drag <Category> to Rows. Tableau makes a bar chart first
3. Click <Show me > select - Pie Chart. Tableau will auto-set:
 - Profit to Angle and Size
 - Category to Color
4. Make it bigger:
 - Option 1: Click <Fit > Entire View, or
 - Option 2: Click <Size> on the Marks card and drag the slider right
5. Drag <Category> to – Label- on the Marks card to show Furniture, Office Supplies, Technology on the slices.

OUTPUT:

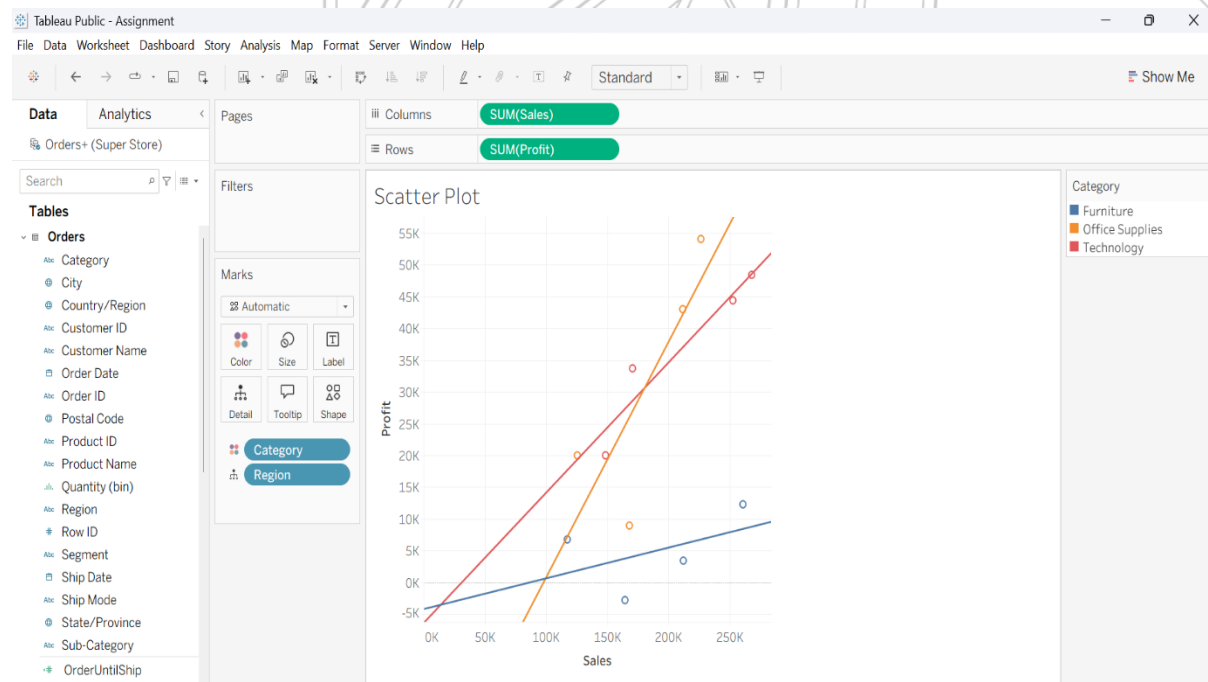


Sheet – 10 - Scatter Plot:

Steps:

1. Drag <Sales> to Columns for the X-axis
2. Drag <Profit> to Rows for the Y-axis. You'll get one mark to start
3. Drag <Category> to - Color - on the Marks card. Each category gets a different color: Furniture, Office Supplies, Technology
4. Drag <Region> to - Detail - on the Marks card. This splits each category into Central, East, South, West, so more marks appear
5. Add trend lines: Open the <Analytics pane > drag - Trend Line - onto the chart select – Linear.

OUTPUT:

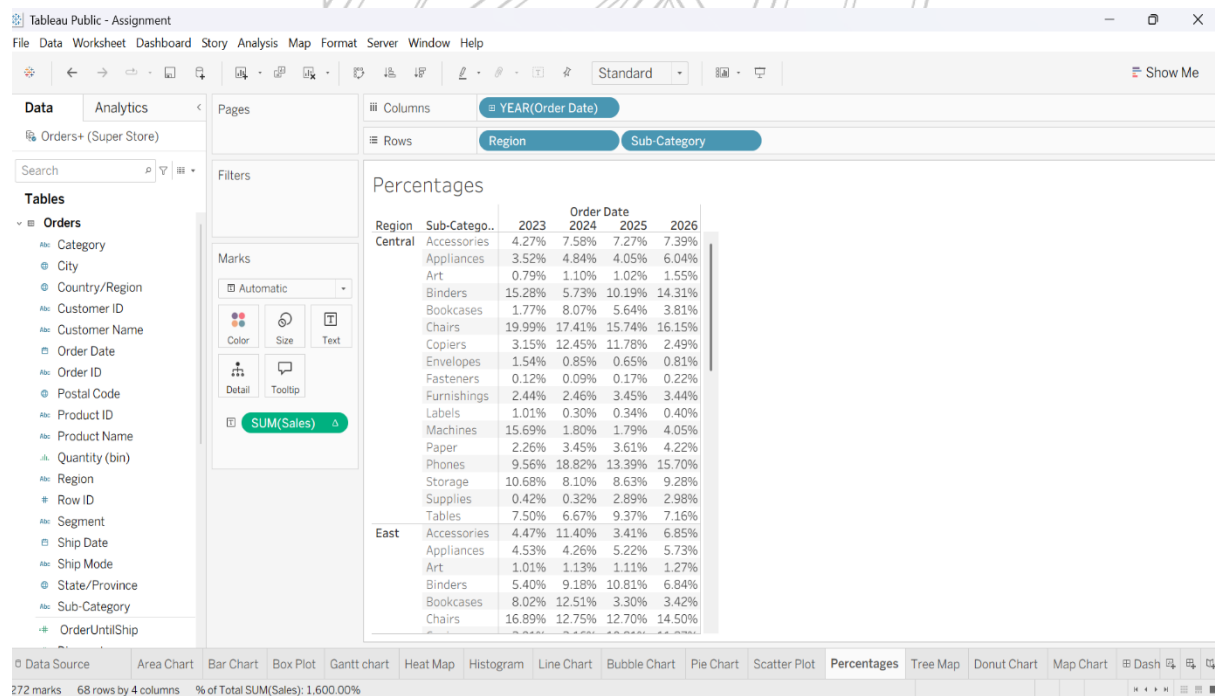


Sheet – 11 – Percentage:

Steps:

1. Drag <Order Date> to Columns. Tableau sets it to - YEAR(Order Date)
2. Drag <Sub-Category> to Rows. Now you have years across the top and sub-categories down the side
3. Drag <Sales> to – Text- on the Marks card. Each cell shows SUM(Sales) for that year + sub-category
4. Drag <Region> to Rows, drop it to the left of Sub-Category. Rows now break down as Region → Sub-Category
5. Right-click - SUM(Sales) on the Marks card > Add Table Calculation
6. Set <Calculation Type> to - Percent of Total -, - Compute Using - to - Pane (Down), then click (OK).

OUTPUT:

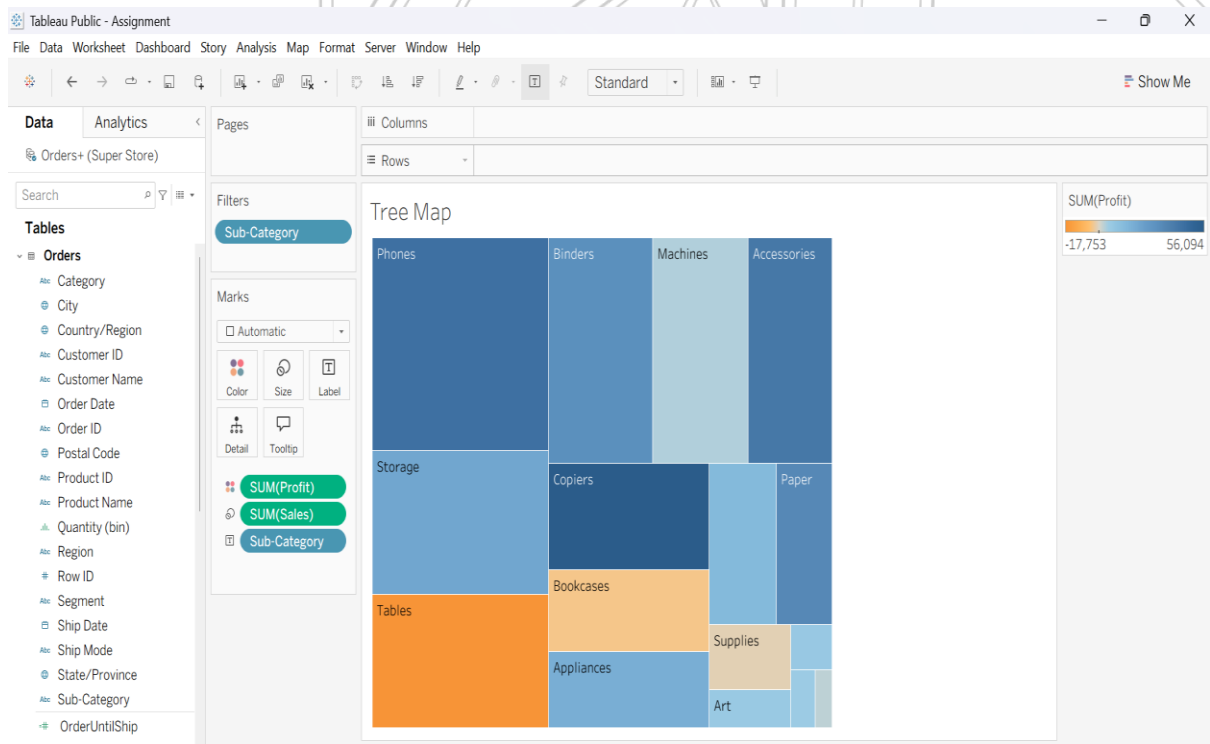


Sheet – 12 – Tree map:

Steps:

1. Drag <Sub-Category> to Columns
2. Drag <Sales> to Rows. Tableau makes a bar chart first
3. Click <Show Me > select – Tree map . Now each sub-category is a rectangle sized by Sales
4. By default: <Size = Sales> and Color = Sales*. Large dark boxes = high sales, small light boxes = low sales
5. Drag <Ship Mode> to – Color - on the Marks card to color rectangles by First Class, Second Class, Standard Class, Same Day. Size still shows Sales
6. Optional: Undo Ship Mode, then drag - Profit - to Color. Now - Size = Sales - and - Color = Profit to spot profitable vs unprofitable sub-categories.

OUTPUT:

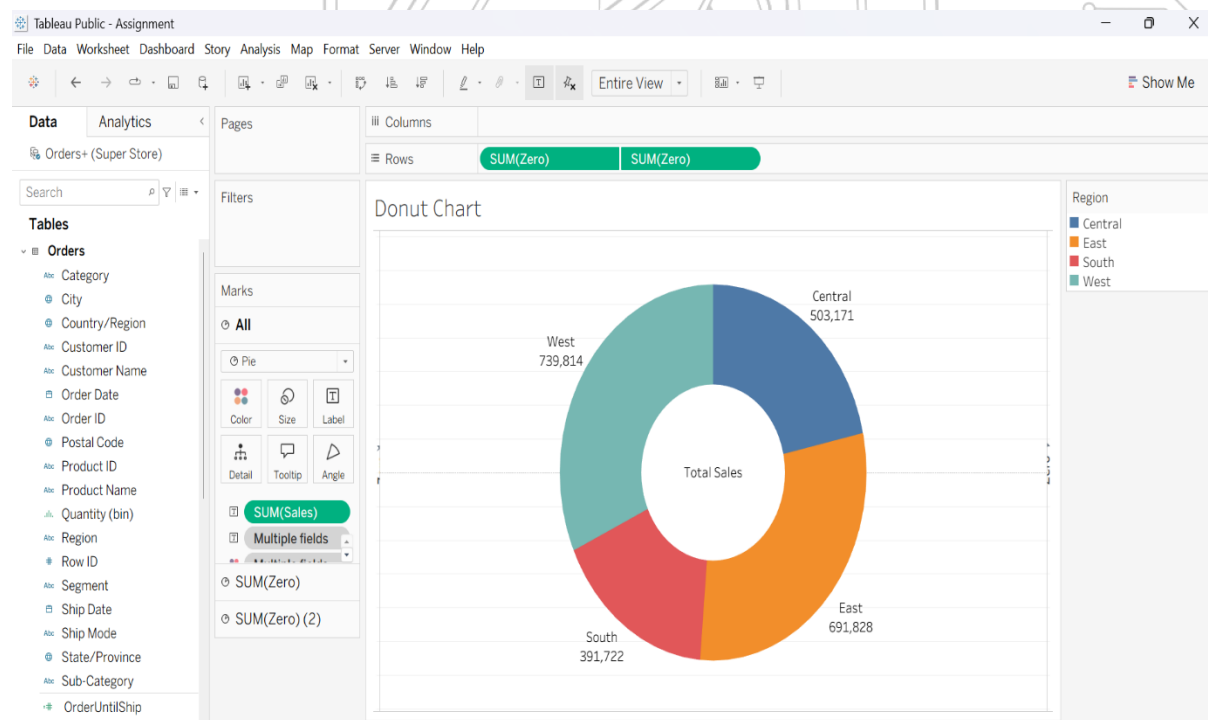


Sheet – 13 - Donut Chart:

Steps:

1. Create a field: Analysis > Create Calculated Field > name it `Zero` > formula is just `0` > OK
2. Make the base pie: Drag `Zero` to Rows twice. Change Marks card to – Pie - and set Fit > Entire View . Increase size with the – Size- slider
3. Make it dual axis: Right-click the second `SUM(Zero)` in Rows > Dual Axis. The pies now overlap
4. Make the hole: Select the second `SUM(Zero)` Marks card > reduce - Size - slightly > set Color to White
5. Add data: On the first `SUM(Zero)` Marks card, drag – Region - to Color and Label, and drag - Sales - to Label and Angle. Slices will show sales by region
6. Add center label: On the second `SUM(Zero)` Marks card, add – Sales - to Label to show total sales in the center.

OUTPUT:



Sheet – 14 – Map Chart:

Step 1: Create a Basic Map From the Data Pane, drag Country to the workspace. Tableau automatically generates a map using Latitude and Longitude.

Step 2: Add Sales Information Drag Sales from Measures to Color on the Marks card. The map now displays sales values with different colors.

Step 3: Change Map Type In the Marks card, click the Automatic drop-down. Select Map or Filled Map. Map → Shows circles on locations. Filled Map → Fills states/countries with colors.

Step 4: Adjust Colors Click Color on the Marks card. Select Edit Colors. Choose a color palette like Purple-Sequential. Check Stepped Color → Set to 5 steps for clear bands. Click OK.

Step 5: Add Labels Drag State to Label on Marks card. Drag Sales to Label. Click Label → Check Allow labels to overlap other marks → OK.

Now each state shows its sales value.

Step 6: Add Region Filter Drag Region to the Filters shelf. Select All → Click OK. Right-click Region pill in Filters → Show Filter.

This allows regional analysis on the map.

Step 7: Add Profit Details Drag Profit to Tooltip on Marks card. Click Tooltip → Edit text to: State: <State> Sales: <SUM(Sales)> Profit: <SUM(Profit)> Hover over any state to view Sales and Profit details

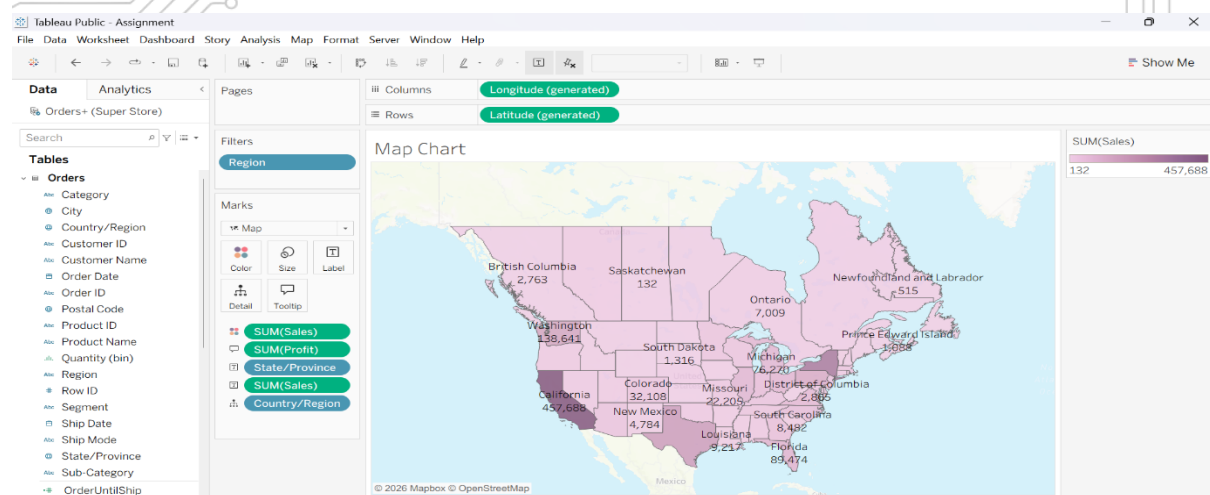
Step 8: Format the Map Go to Map → Map Layers. Adjust: Background → Style: Outdoors Washout → Set to 70% Map Layers → Check Country/Region Names and State/Province Names Bonus Steps: Extra polish

9. Add a Title: Double-click Sheet title → Type Sales by State - Filled Map

10. Fix unknown locations: If you see a grey box, click it → Edit Locations → Match to correct country/state

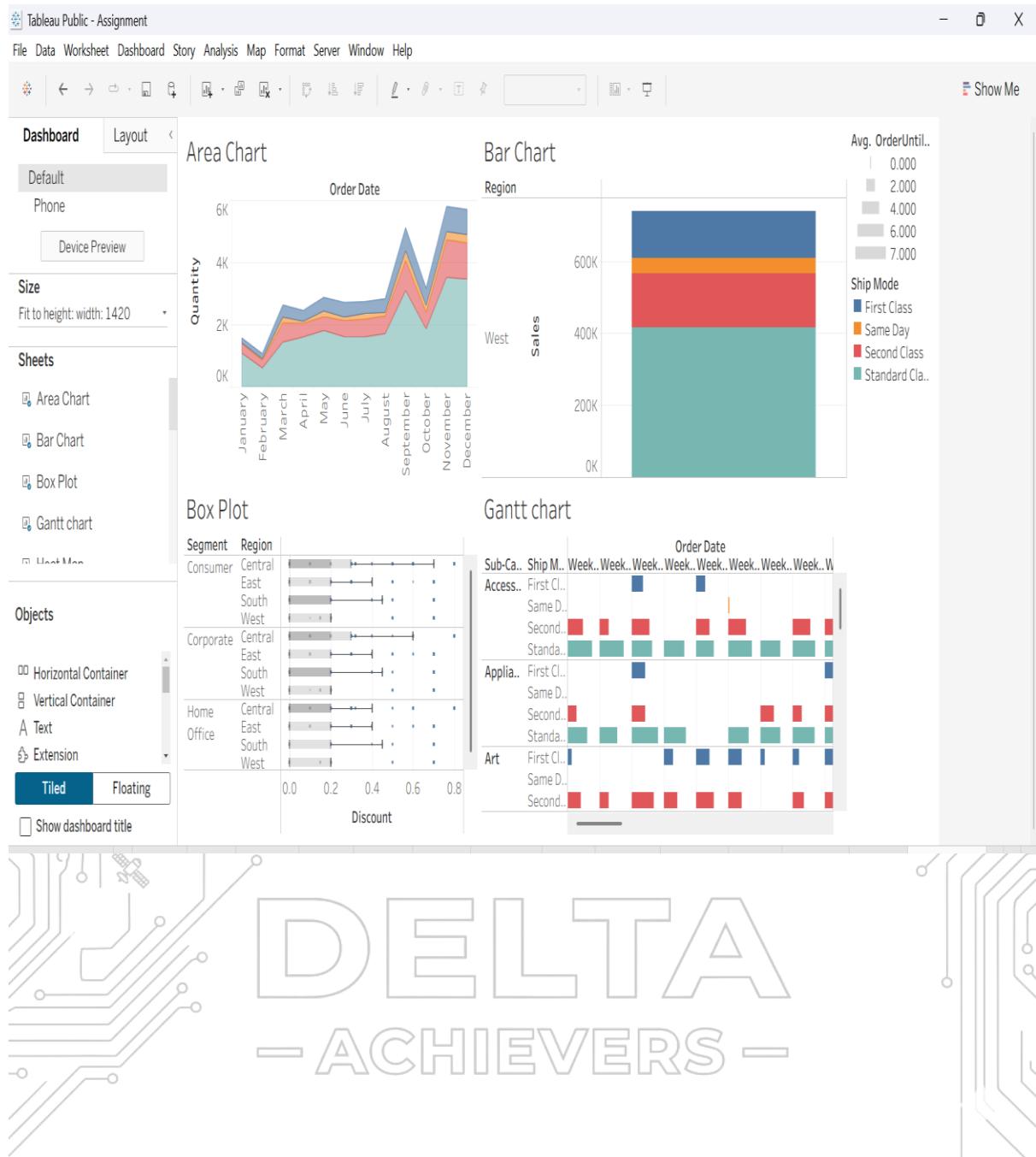
11. Set size for symbol maps: If using Map instead of Filled Map, drag Sales to Size too. Circles get bigger with higher sales.

OUTPUT:

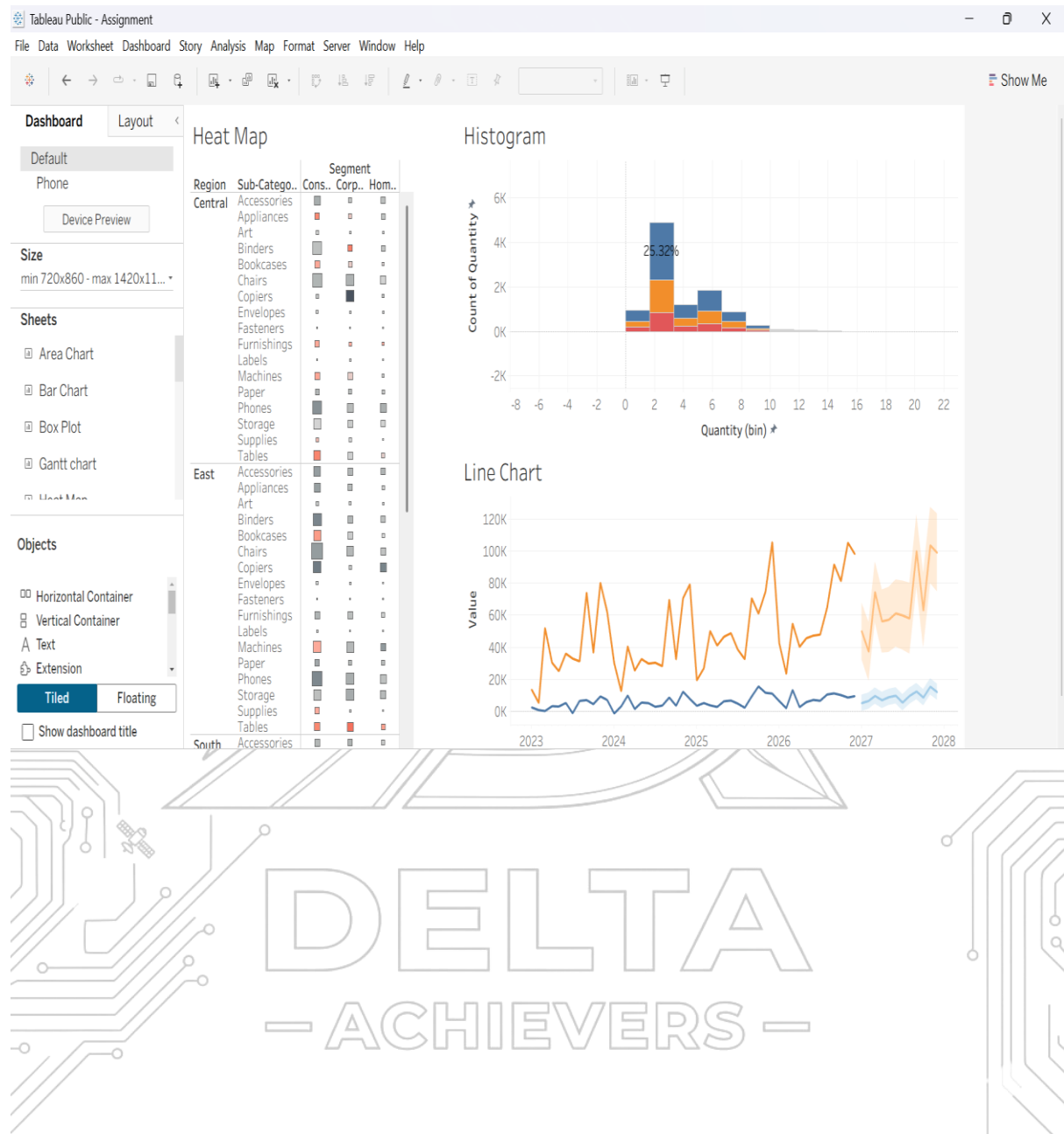


DASHBOARDS

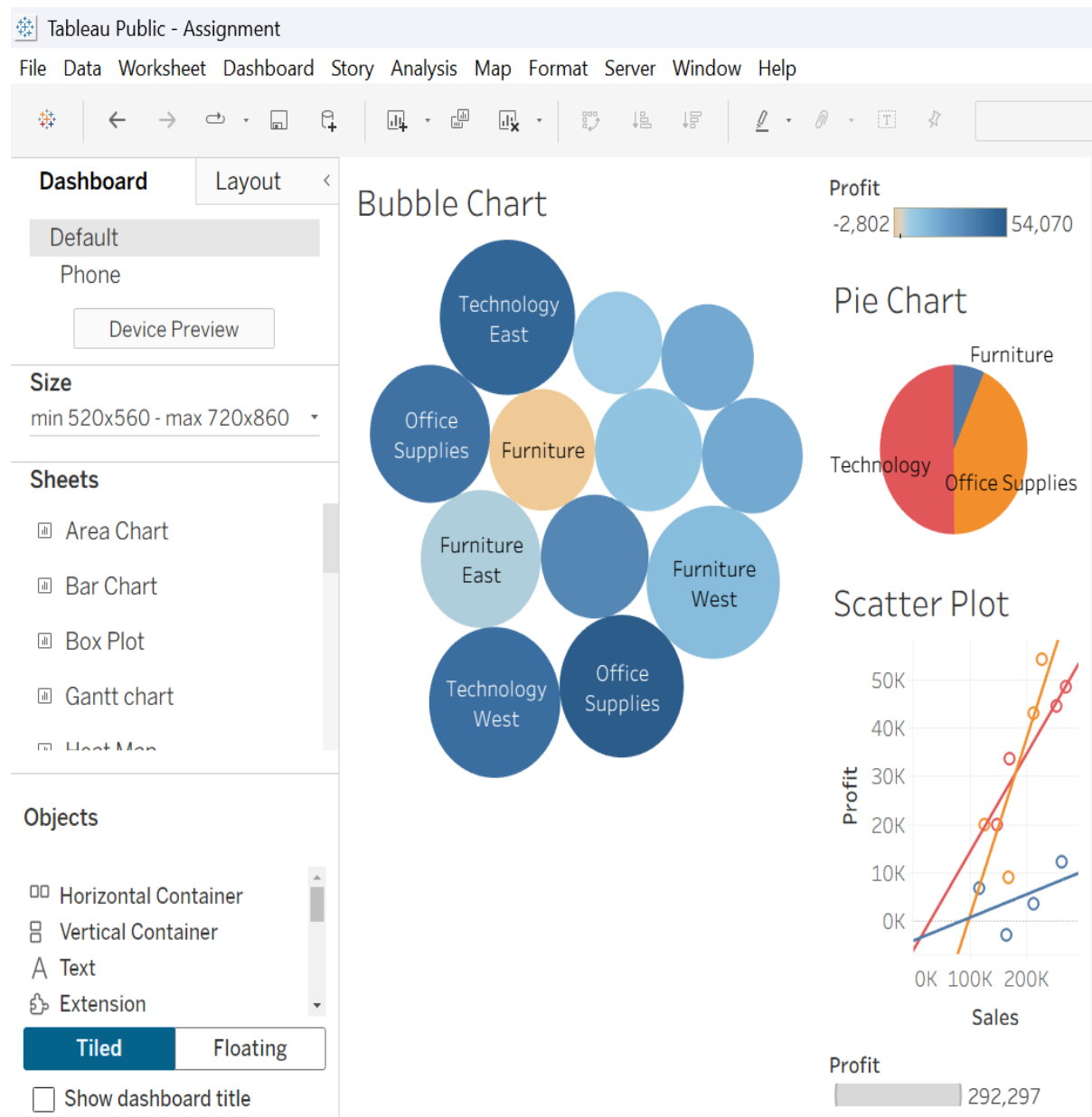
DASHBOARD-01:



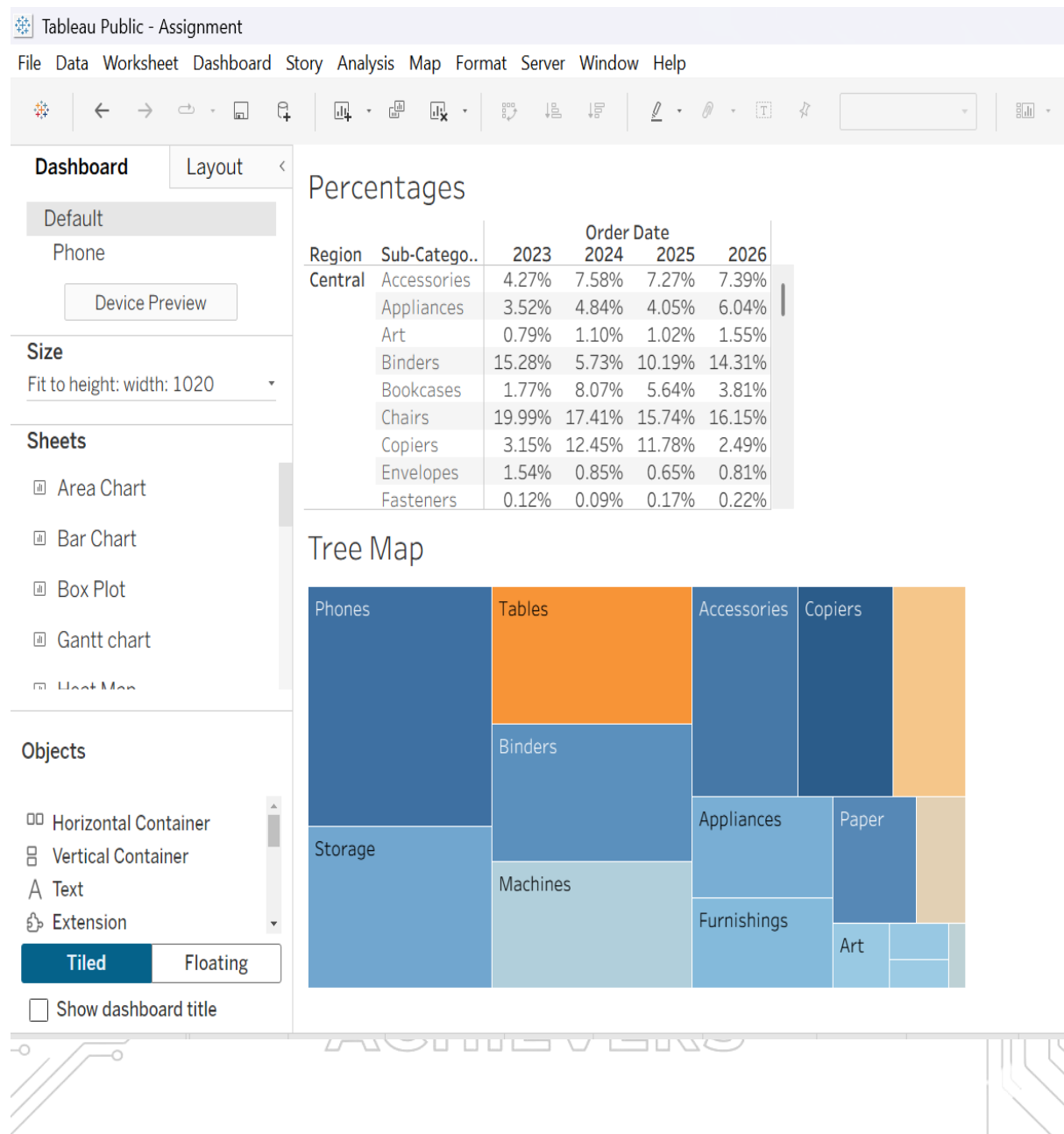
DASHBOARD – 02 :



DASHBOARD – 03:



DASHBOARD – 04:



DASHBOARD – 05:

